

Something About Sea Ice

by
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Abstract

Roughly 200 words for an abstract

Acknowledgments

Say nice things to people here.

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Chapter 1

Introduction

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Figure 1.1: This is where you write a figure caption.

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Chapter 2

Data

Definitely need this chapter if you're using any data from any sources.

2.1 This is how you do sections

2.1.1 And this is a subsection

Then you can refer to a figure in the text, like Figure ??.

Location	Track	Cycle	Seg	Laser	Dates	Year
Chukchi	0381	14/15	05	GT3L	16 Jan/17 April	2022
Chukchi	0373	14/15	03	GT2L	15 Jan/16 April	2022
Chukchi	0373	14/15	03	GT3L	15 Jan/16 April	2022
Chukchi	0815	14/15	03	GT1L	13 Feb/15 May	2022
Chukchi	0815	14/15	03	GT2L	13 Feb/15 May	2022
Beaufort	0381	14/15	05	GT1L	16 Jan/17 April	2022
Beaufort	0320	14/15	05	GT3L	12 Jan/13 April	2022

Table 2.1: This is what tables look like



Figure 2.1: This is a figure caption. That is a penguin.

Chapter 3

Conclusions

Another example section.

Chapter 4

Future Work

This is a section that is mostly helpful as a place to put the ideas you don't have time to pursue.

Chapter 5

How to write things in LaTeX

5.1 Welcome to the World of LaTeX

LaTeX is a powerful typesetting system widely used in academia, particularly in fields like mathematics, computer science, and engineering. Unlike word processors that focus on WYSIWYG (What You See Is What You Get), LaTeX operates on a WYSIWYM (What You See Is What You Mean) principle. You write the content and structure of your document, and LaTeX handles the formatting, resulting in beautifully typeset documents.

5.1.1 Basic Structure

A LaTeX document typically begins with a preamble, where you define the document class and load packages. The main content follows within the `document` environment.

```
\documentclass{article}
\usepackage{amsmath} % Example package

\begin{document}

% Your content here

\end{document}
```

- `\documentclass{article}`: Specifies the document type (e.g., article, report, book).

- `\usepackage{amsmath}`: Loads the `amsmath` package for advanced mathematical typesetting.
- `\begin{document}` and `\end{document}`: Enclose the main content of your document.

Your thesis is written as a book. Also, we've set up the preamble so unless you want to do something weird, you don't need to worry about it.

5.1.2 Text and Paragraphs

To write text, simply type it. LaTeX automatically handles line breaks and paragraph formatting. A blank line creates a new paragraph.

This is the first paragraph.

This is the second paragraph.

This is the first paragraph.

This is the second paragraph.

5.1.3 Sections and Subsections

Organizing your document is crucial. LaTeX provides commands for creating sections, subsections, and subsubsections.

```
\section{Introduction}
```

```
\subsection{Background}
```

```
\subsubsection{Previous Work}
```

Note: a `*` between the section keyword and the bracket makes it not numbered.

```
\subsubsection*{Future Work}
```

5.2 Introduction

5.2.1 Background

Previous Work

Future Work

5.2.2 References

To cite a paper, you will need the reference information to be in the .bib file (bibliography.bib here) and then you do this:

```
\citep{louis2018testing}. If you want to cite more than one, use \citep{louis2018tes
```

That looks like this: [?]. If you want to cite more than one, use [??]. You can also do an in-text citation with ?.

5.2.3 Mathematics

LaTeX excels at typesetting mathematical equations. Inline math is enclosed in single dollar signs ($\dots$), while displayed equations are enclosed in double dollar signs ($\dots$) or the `equation` environment.

The Pythagorean theorem is $a^2 + b^2 = c^2$.

```
\begin{equation}
  E = mc^2
\end{equation}
```

which looks like this: The Pythagorean theorem is $a^2 + b^2 = c^2$.

$$E = mc^2 \tag{5.1}$$

5.2.4 Lists

LaTeX offers environments for creating bulleted and numbered lists.

- `itemize` for bulleted lists.

- `enumerate` for numbered lists.

```
\begin{itemize}
  \item Item 1
  \item Item 2
\end{itemize}
```

```
\begin{enumerate}
  \item First item
  \item Second item
\end{enumerate}
```

5.2.5 Including Images

To include images, use the `graphicx` package and the `includegraphics` command.



Figure 5.1: A sample figure caption.

```
\usepackage{graphicx} % Already added to preamble

\begin{figure}[htbp]
  \centering
  \includegraphics[width=0.5\textwidth]{image.png}
  \caption{A sample image.}
  \label{fig:sample}
\end{figure}
```

5.2.6 Tables

The `tabular` environment is used to create tables. Note there are tools on the internet that can convert an excel formatted table to a latex one.

```
\begin{tabular}{|c|c|c|}
  \hline
  Header 1 & Header 2 & Header 3 \\
  \hline
  Data 1 & Data 2 & Data 3 \\
  \hline
\end{tabular}
```

Header 1	Header 2	Header 3
Data 1	Data 2	Data 3

Happy typesetting!

Appendix A

Seasonal Ice Records

A.1 An Appendix

This is where you put information and data that isn't core to your explanation, but that you want to not lose.

Also this is what a table looks like:

Date	Location	Recorded Ice Behavior
2022-06-30	Utqiaġvik (Barrow)	Pack ice near shore
2022-06-29	Utqiaġvik (Barrow)	Break-out of shorefast ice
2022-06-29	Utqiaġvik (Barrow)	Pieces of shorefast ice breaking off/Some shorefast ice remaining
2022-06-27	Utqiaġvik (Barrow)	Breakup of shorefast ice
2022-06-27	Utqiaġvik (Barrow)	Pack ice visible from shore, Some shorefast ice remaining
2022-06-24	Utqiaġvik (Barrow)	Shorefast ice is deteriorating
2022-06-21	Utqiaġvik (Barrow)	Breakup of shorefast ice
2022-06-20	Utqiaġvik (Barrow)	Break-out of shorefast ice
2022-06-17	Utqiaġvik (Barrow)	Pack ice near shore
2022-06-16	Utqiaġvik (Barrow)	Pieces of shorefast ice breaking off
2022-06-15	Utqiaġvik (Barrow)	Pieces of shorefast ice breaking off
2022-06-14	Utqiaġvik (Barrow)	Shorefast ice is deteriorating
2022-06-12	Utqiaġvik (Barrow)	Pack ice adding to shorefast ice
2022-06-10	Utqiaġvik (Barrow)	Pack ice near shore
2022-05-31	Utqiaġvik (Barrow)	Lead open
2022-05-17	Utqiaġvik (Barrow)	Lead open
2022-05-16	Utqiaġvik (Barrow)	Pack ice all packed in
2022-05-14	Utqiaġvik (Barrow)	Pack ice all packed in
2022-05-12	Utqiaġvik (Barrow)	Pack ice all packed in
2022-05-11	Utqiaġvik (Barrow)	Pack ice is moving
2022-04-19	Utqiaġvik (Barrow)	Lead open
2022-04-02	Utqiaġvik (Barrow)	Lead open
2022-02-20	Gravel Pit	Lead open
2022-01-24	Gravel Pit	Lead open
2022-01-18	Gravel Pit	First stable ice for travel
2021-12-15	Piqniq (Duck Camp)	"Ice ridge/s, Shorefast ice is grounded"
2021-12-13	Piqniq (Duck Camp)	Shorefast ice is rough
2021-11-30	Utqiaġvik (Barrow)	Ice ridge/s, Shorefast ice is thickening
2021-11-18	Niksiuraq	Young ice freezing in place
2021-11-15	Piqniq (Duck Camp)	Stationary ice along shore
2021-10-22	Utqiaġvik (Barrow)	Start freeze up on ocean

Table A.1: A summary of observations from the Alaska Arctic Observatory and Knowledge Hub from the 2021-2022 winter season